Risk aware and explainable fabric defect detection

Individual ANN coursework

Questions 1 to 6

P.M.P Senevirathna

COMSCDS252P-002

Risk aware and explainable fabric defect detection

Individual artificial neural network coursework using RAW-FABRID V1
Questions 1 to 6

The application helps a textile Quality Assurance Inspector examine a 256 x 256 fabric patch. It shows the trained model's score, its normal or defective decision, and a Grad-CAM heatmap that guides visual inspection. The inspector decides whether to release or hold the fabric. The application does not certify a whole roll or control production machinery.

The approved dataset was downloaded from its publisher and checked before modelling. The publisher's training folder contains only normal patches, so the labelled patches were combined and assigned to a fixed supervised training, validation and test split. This assignment kept the publisher's fabric origin groups, source images and duplicate relationships together. The results therefore describe this supervised protocol, rather than the publisher's original benchmark for anomaly detection.

Evidence	Completed result
Full release	19,852 patches, including 687 defective. The publisher's SHA-256 checksum was verified
Training search	21 ANN configurations, 28 classical configurations including four pilots, and three seeds for the selected model
Final population	All 2,996 held out patches, including 96 defective, used in the same ANN, SVM and VLM comparison
Scientific failures	Four actual unsuccessful attempts, including two substantive approaches
XAI	Three genuine incorrect predictions with measured top, low and random occlusion checks
Engineering checks	52 tests passed, along with actual HTTP and browser checks and four notebooks
Individual presentation	The individual live defence must be completed separately

The final artificial neural network (ANN), ann-search-018, has 1,518,881 parameters and achieved a test F2 score of 0.6883. The tuned support vector machine (SVM), svm-search-042, achieved 0.0623. The final ANN performed better in this comparison. No model choice changed after the test scores were seen.

The study aims to provide results that can be reproduced and explained. It reports prediction weaknesses, limited coverage of fabric origins, uncalibrated model scores and unfavourable explanation checks. The system remains a research prototype and does not guarantee production quality. After evaluation files were lost, the same fixed evaluation was physically computed a second time. This exception to the brief's rule requiring one evaluation is disclosed, and lecturer acceptance has not been verified.

Q1 Dataset origin and prediction target

Where the data came from and what was checked

Ruben Perez-Llorens, Teresa Albero-Albero and Javier Silvestre-Blanes published RAW-FABRID V1 on 1 April 2026 under CC BY 4.0, with DOI 10.17632/db6g85xsyg.1 [1]. They describe the images as industrial grayscale imagery from line scanning. This implementation follows the downloaded files as audited, rather than relying only on the website description.

Release object	Audited count / shape
Original images	709 images at 1792 x 1024 pixels, stored as 8 bit grayscale PNG
Original masks	204
All image patches	19,852 grayscale patches of 256 x 256 pixels
Publisher train/good	14,196 normal patches only
Publisher test/good	4,969 normal patches
Publisher test/defect and patch masks	687 defective patches and 687 corresponding masks

Label 1, meaning defective, is assigned when a patch's annotation mask contains at least one completely white defect pixel with value 255. Label 0 requires an explicit normal classification in the publisher's metadata or folder structure. A normal patch stays normal even if a defect appears elsewhere in its larger source image. A missing mask is never assumed to mean normal. The classifier does not receive masks, source identifiers or file paths that could reveal the label.

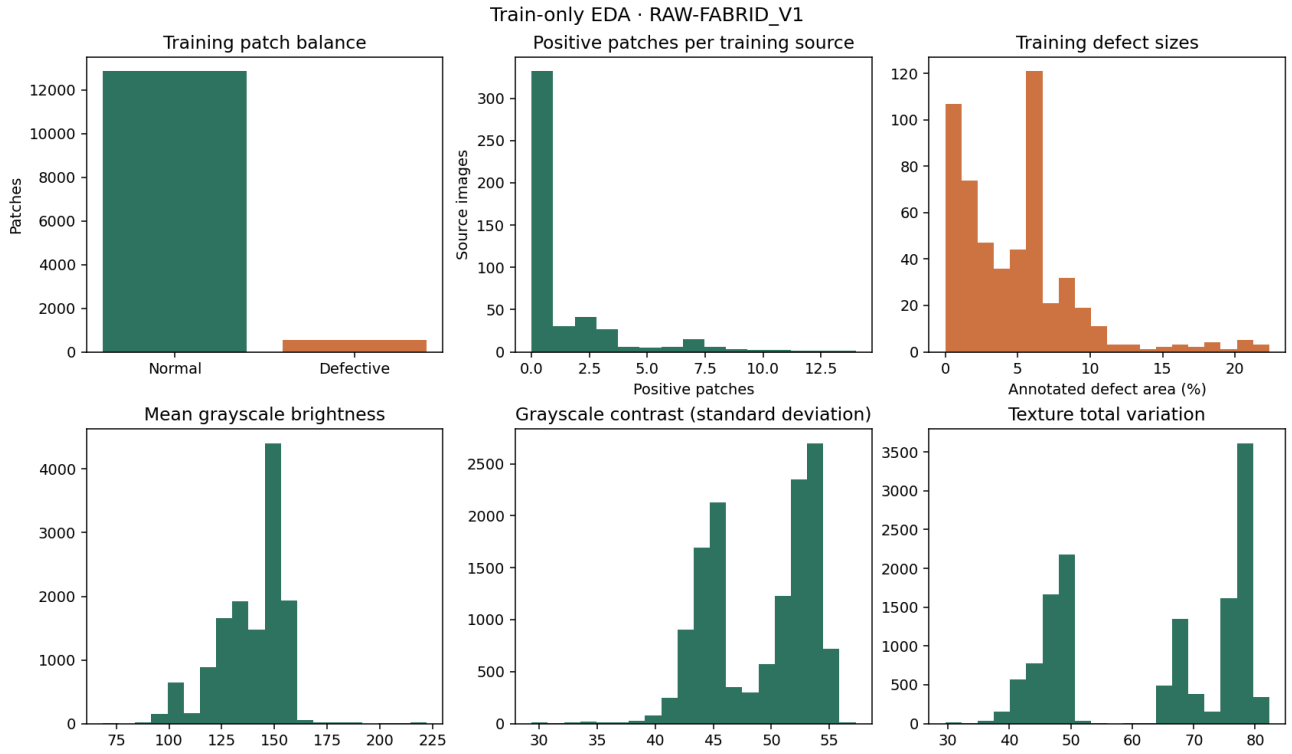
Each patch is traced through the patch CSV, where patch_name identifies the patch and parent_raw identifies its original image. The CSV for high resolution images links filename to roll_id. The publisher's README describes roll_id as an anonymised fabric origin. Each origin is kept together as a precaution, although the available information does not establish that each identifier represents exactly one physical roll, fabric type or manufacturer.

Intermediate gray values were found around the edges of 471 patch masks, although a strict reading of the publisher's description suggests binary masks with two levels. Counting only pixels with value 255 reproduces the recorded defect area for all 687 defective patches. The original mask bytes are preserved. Alignment between images and their masks is checked, and patch labels are verified, before the data manifest is fixed.

The complete archive contains 4,133,046,386 bytes. Its verified SHA-256 checksum is f6b4f82cce2f0220bbab3968a74b5337ebf2bbfd5e33d0548d34e1ebf80f85b9. Because the archive exceeds 500 MB, the submission includes a script that reproduces the download using byte ranges and an authentic sample of 168 patches. Defects are deliberately more common in that sample to support software checks. The sample is never used to calculate assessment results.

Q1 Exploring the training images

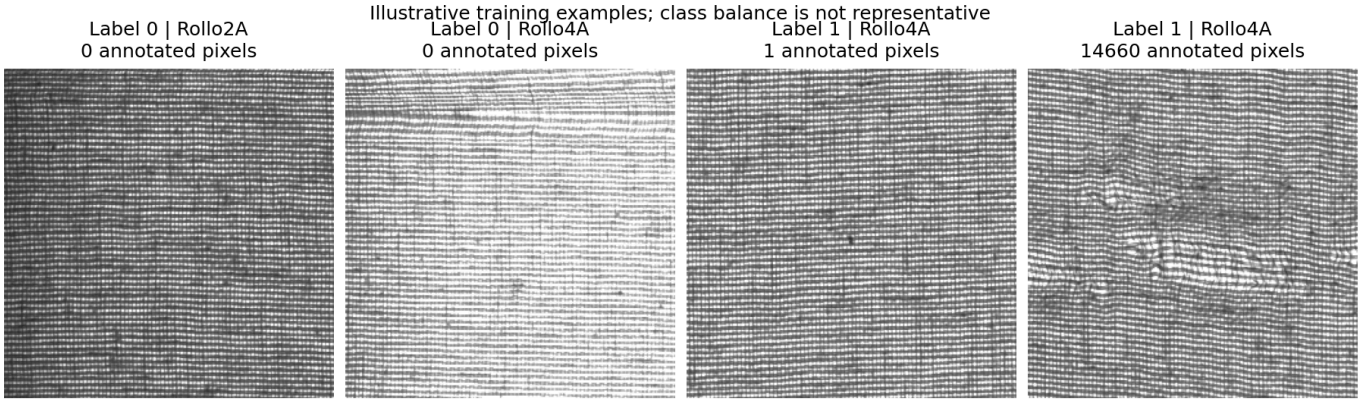
Image characteristics and the size of annotated defects



Training image statistics and annotation sizes

The training set contains 13,384 patches, including 539 defective patches, or 4.03%. They come from 478 source images but only four groups defined by the publisher's origin field. Defective patches occur in 146 source images. Patches from the same source are related and should not be treated as independent observations.

Among defective training patches, the median annotated defect area is 4.72% of the patch. The lower and upper quartiles are 1.39% and 6.04%. The smallest annotation covers one pixel, or 0.0015%, and the largest covers 22.37%. The label definition therefore includes defects that may be too small for reliable human recognition or a convolutional neural network (CNN) with coarse spatial features.



Examples from the training set

Mean patch intensity ranges from 68.6 to 222.1. Contrast, measured as the standard deviation of intensity, ranges from 29.4 to 57.3. These values describe image acquisition and do not establish verified defect categories. All exploratory data analysis (EDA) used to guide modelling was restricted to the training set.

Q1 Preventing data leakage and choosing evaluation measures

One fixed supervised split and evaluation procedure

Partition / publisher origins	Patches	Positive	Sources
Train: Rollo2A, 4A, 5A, 7A	13,384	539	478
Validation: Rollo6A	3,472	52	124
Test: Rollo1A, 3A	2,996	96	107

The split was created with seed 42. Of 128 candidate draws checked for structural integrity and the presence of both classes, draw 32 was selected without consulting model performance. Hashes of decoded pixels found no exact duplicates. The near duplicate audit considered 324,109 candidate pairs, performed 7,587 checks comparing all pixels and found no confirmed matches within a mean absolute error (MAE) of 3/255. No connected source, origin or duplicate group crosses a partition. The checks can still miss rotations, different crops and relationships that the publisher has not documented.

A false negative misses a defective patch and may waste material later in production. A false positive creates an unnecessary inspection or hold. These are relative consequences, rather than measured factory costs. The main evaluation measure is defect class $F2 = \frac{5\,TP}{5\,TP + FP + 4\,FN}$. Here TP means correctly detected defects, FP means false alarms and FN means missed defects. F2 gives more weight to recall while still penalising poor precision. Accuracy alone can be misleading because most patches are normal. The report also gives precision, recall, F1, false positive rate, confusion counts, average precision (AP) and area under the receiver operating characteristic curve (ROC-AUC) when continuous scores are available.

Every model that produces a score uses the same complete validation set to choose its decision cutoff. The search considers 101 fixed score quantiles and the extremes, selecting the cutoff with the highest F2. Ties favour higher precision and then the higher cutoff. An unresolved vision language model (VLM) output has value -1. It remains in the evaluation denominator and counts as an error for its true class. These labels are never treated as probabilities, and ranking measures are omitted for VLM outputs that contain only labels.

The exploratory analysis led to three modelling requirements. Rare defects call for evaluation that emphasises recall and for a candidate using weighted loss whose weight comes from training data. Tiny defects require the full 256 x 256 detail, without cropping or blur, together with spatial explanation checks. Having only four training origins calls for modest model capacity, a comparison using transfer learning and cautious evaluation by independent groups. Defect prevalence also differs between training and validation, so a gap between their loss values cannot by itself establish overfitting.

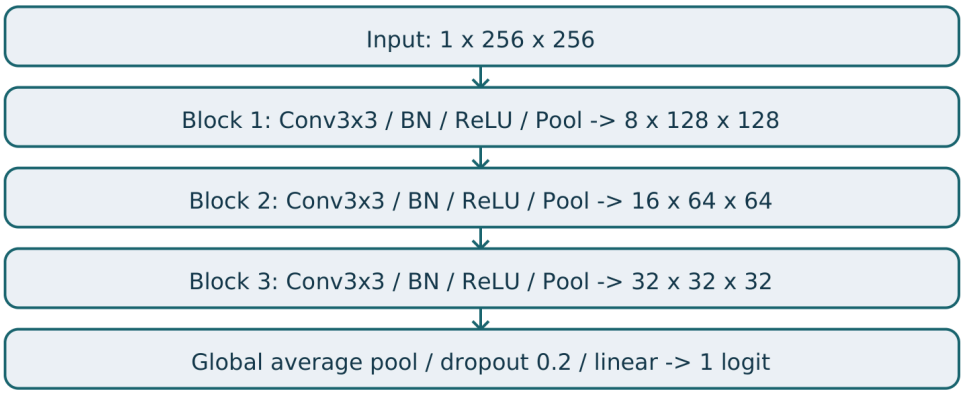
The split's SHA-256 checksum is af7daf13d0854ceeba004f1a06182bac778ac716021dec3d8fc40edf265aa76f. Model choices, checkpoints, prompts and code were fixed before any access to the test set. A later runtime reset lost the first completed test files. They were reconstructed by computing the identical protocol a second time, without retraining, selecting again or changing cutoffs. This is one benchmark population, rather than independent repeated validation. The original fixed plan records the intended single computation, while the recovery record discloses the actual two computations. Only two test origins are available, which limits uncertainty estimates.

Q2a The initial neural network

Initial model ann-search-000

The initial model has 5,977 parameters, all trainable. The complete table of individual layers was measured using model hooks and is saved in reports/ann-search-000-architecture.csv. The table below groups those measured modules and omits the batch dimension. The model outputs one unrestricted defect logit, or raw score. A sigmoid function converts it into the score shown in the application and used to choose the validation cutoff.

A single channel 1 x 256 x 256 image passes through three convolution, batch normalisation, ReLU and pooling blocks. Their outputs are 8 x 128 x 128, 16 x 64 x 64 and 32 x 32 x 32. Global average pooling, dropout 0.2 and a linear layer then produce one logit.



Structure of the initial CNN

Stage	Operation	Output shape	Parameters
Block 1	3x3 Conv, BN, ReLU, 2x2 max pool	8 x 128 x 128	88
Block 2	3x3 Conv, BN, ReLU, 2x2 max pool	16 x 64 x 64	1184
Block 3	3x3 Conv, BN, ReLU, 2x2 max pool	32 x 32 x 32	4672
Global average pool	AdaptiveAvgPool2d(1)	32 x 1 x 1	0
Classifier	Flatten, dropout 0.2, Linear(32,1)	1	33

Three convolutional blocks were used on the assumption that useful local weave patterns can occur in different image positions. Starting with width 8 limits capacity when there are few independent origins. Widths of 8/16/32 allow later blocks to combine increasingly complex texture features. Same padding preserves edge information before pooling. Batch normalisation stabilises activation scales, and ReLU lets the model learn nonlinear relationships. Pooling and global averaging reduce computation and parameter count but may dilute very small defects. Dropout 0.2 regularises the compact classifier.

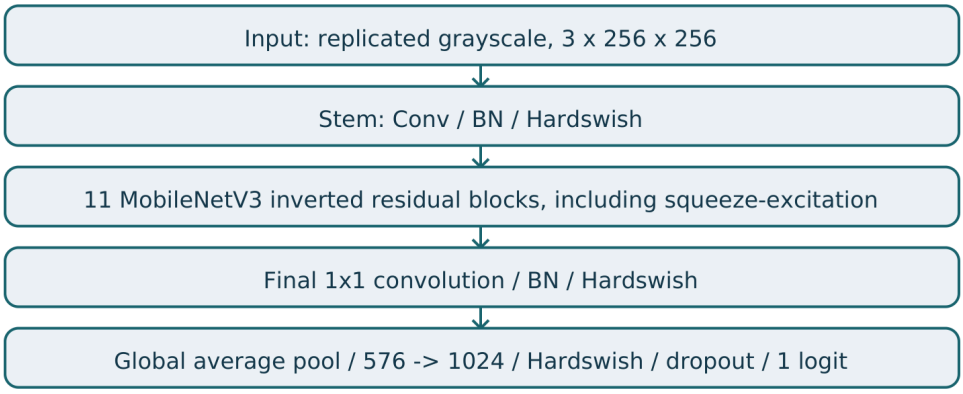
BCEWithLogitsLoss was used because it provides a numerically stable loss for a single binary logit. Its initial defect class weight is (13,384-539)/539 = 23.8312, calculated only from training data. This addresses class imbalance but does not necessarily calibrate scores or represent the business cost ratio. The initial model uses AdamW with learning rate 0.001, batch size 32 and four screening epochs. Horizontal and vertical flips are applied only during training. Random seeds are fixed. Unsupported deterministic kernels produce warnings, so complete reproducibility is not guaranteed without evidence.

Q2a The selected final neural network

Selected model ann-search-018

The selected model has 1,518,881 parameters, all trainable. Its complete table of individual layers was measured using model hooks and is saved in reports/ann-search-018-architecture.csv. The table below groups those modules and omits the batch dimension. The single output is an unrestricted defect logit. A sigmoid function converts it into a display score and supports threshold selection using validation data.

The grayscale image was copied into three channels, giving an input of 3 x 256 x 256. A convolution, batch normalisation and Hardswish stem feeds 11 MobileNetV3 inverted residual blocks, including squeeze and excitation. A final 1x1 convolution, batch normalisation and Hardswish are followed by global average pooling. The classifier maps 576 features to 1024, applies Hardswish and dropout, and outputs one logit.



Structure of the selected MobileNetV3 model

Stage	Operation	Output shape	Parameters
Features 0	Conv3x3 / BN / HSw	16 x 128 x 128	464
Features 1	IR / BN / ReLU / SE	16 x 64 x 64	744
Features 2	IR / BN / ReLU	24 x 32 x 32	3864
Features 3	IR / BN / ReLU	24 x 32 x 32	5416
Features 4	IR / BN / HSw / SE	40 x 16 x 16	13736
Features 5	IR / BN / HSw / SE	40 x 16 x 16	57264
Features 6	IR / BN / HSw / SE	40 x 16 x 16	57264
Features 7	IR / BN / HSw / SE	48 x 16 x 16	21968
Features 8	IR / BN / HSw / SE	48 x 16 x 16	29800
Features 9	IR / BN / HSw / SE	96 x 8 x 8	91848
Features 10	IR / BN / HSw / SE	96 x 8 x 8	294096
Features 11	IR / BN / HSw / SE	96 x 8 x 8	294096
Features 12	Conv1x1 / BN / HSw	576 x 8 x 8	56448
Pool/classifier	GAP, Linear 576 to 1024, Hardswish, dropout 0.2, Linear 1024 to 1	1	591873

The 21 candidates were ranked by validation F2, then AP and finally fewer parameters. The selected configuration is {'name': 'mobilenet'}, using AdamW at learning rate 0.0003 and batch size 32. Augmentation is disabled and weighted loss is enabled. Epoch 2 of 4 supplies the selected checkpoint. The separate 12-epoch refinement was a candidate declared in advance and did not automatically replace the best screened model.

Q2a Reasons for the final architecture

What each component contributes and where it can fail

Depth experiments were used to test how features build across successive receptive fields, while width experiments were used to test model capacity. The custom CNN normalises input using (gray/255-0.5)/0.5. MobileNet copies grayscale values into three channels and uses ImageNet means and standard deviations. Every layer is fine tuned from genuine ImageNet weights [2,3], with the original classifier replaced by one output. Images are neither resized nor centre cropped. Transfer learning offers a way to work with limited data, although its original image domain differs from industrial fabric and its features can be too coarse.

In the architecture table, IR means inverted residual. These blocks expand features where configured, use depthwise convolution and project them to the output channels. They add a residual connection when the shapes match. BN means batch normalisation, and HSw means Hardswish. Squeeze and excitation (SE) uses ReLU and Hardsigmoid to control channel contributions. Global average pooling (GAP) and dropout 0.2 limit the classifier head's capacity. Batch normalisation statistics come only from training data. Prediction and Grad-CAM run in evaluation mode. The CSV files for individual layers provide the full detail.

Component	Reason for retaining it	Relevant limitation
Replicated grayscale / ImageNet normalisation	Matches the genuine pretrained input with three channels without changing patch geometry.	Colour pretraining differs from industrial grayscale. Replication adds no colour information.
Depthwise convolution and pointwise projection	Separates spatial filtering from channel mixing with fewer parameters than a similarly wide dense convolution.	Progressive stride reduces detail of tiny defects. The final feature map is coarse.
Inverted residual / shortcut between matching shapes	Expanded features support local combinations. Shortcuts preserve information and gradient flow when dimensions match.	Widths/strides are inherited from MobileNetV3, not separately optimised for this dataset.
BatchNorm	Reuses trainable scale/offset and running statistics estimated during training. Eval mode keeps serving consistent.	Origin or illumination shift can invalidate the training statistics.
ReLU / Hardswish / SE gates	ReLU supplies nonlinearity. Hardswish is x*ReLU6(x+3)/6. SE pools channel context and uses Hardsigmoid gates, as in the transferred architecture.	Channel/global context can favour weave background. Positive CAM attribution does not prove defect causality.
Global average pool	Converts spatial features into a fixed channel vector without a large flattened classifier.	Small defect responses can be diluted by averaging.
576 to 1024 to 1 head. Dropout 0.2	Retains the pretrained nonlinear head structure and replaces the ImageNet output with one binary logit. Dropout regularises training only.	The 1024-unit head width is inherited, not a demonstrated optimum. All layers are fine tuned.

The final model keeps BCEWithLogitsLoss and the defect class weight calculated from training data. Its learning rate, augmentation setting and selected epoch were chosen on validation results. Applying the fixed cutoff to the sigmoid score produces the class decision. That score is not calibrated confidence. The full layer table and saved configuration allow the design to be inspected.

Q2b Alternative methods and expectations before testing

When other methods would be suitable

Alternative	When appropriate	Why retain or reject here
LBP/HOG with SVM	Limited labelled data. Patches dominated by texture	Retained and tuned. Spatial/directional features may generalise better than learned filters.
Small scratch CNN	Local cues specific to the task, enough independent coverage	Retained as initial model. Explicit depth/width help defend design. Only four origins limit learning.
Pretrained compact CNN	Limited labels and reusable visual features	Genuine fine tuning tested. Domain mismatch, grayscale and tiny defects can weaken transfer.
Vision transformer	Large diverse data or suitable domain pretraining	Rejected as a primary candidate: compute/data requirements and tradeoffs involving local detail exceed the scope of this study.
Anomaly detection using only normal patches	Representative normal process, rare unknown defects	Publisher split supports it, but it would change the approved supervised comparison and score semantics.
Supervised segmentation	Reliable dense labels and need for exact defect boundaries	Potential next study. Binary patch classification is the approved task. Grad-CAM is not segmentation.
RNN/LSTM/GRU	True temporal sensor/frame sequences	Rejected: patch pixels are spatial, not a production sequence ordered in time.

An ANN was considered suitable in principle because images contain spatial texture patterns whose combinations may reveal defects. Its practical value still needs to be demonstrated against a strong baseline built from image descriptors. Working with images does not by itself guarantee a better F2 score.

Before final testing, docs/pre-evaluation-hypothesis.md recorded the expectation that a fine tuned compact CNN might exceed the SVM on F2 if coverage of independent sources was sufficient. Adaptable local filters and transferred visual features supported that expectation. Tiny defects diluted by pooling, regular weave, the difference between colour pretraining and grayscale fabric, and few independent origins were reasons the classical model could instead win. The final results do not change that earlier expectation.

Q3a Study design and experiment records

Model selection uses only training and validation data

Family	Measured budget / search space
ANN screening	20 configurations, up to four epochs each, early stopping patience 3, seed 42
ANN refinement	Best screened configuration retrained for up to 12 epochs. Remains one candidate
Seed repetitions	Selected configuration, seeds 42 / 123 / 2026. No seed chosen by test score
Main classical search	24 configurations: 2 feature sets x 2 kernels x 3 C values x 2 class weights
Classical pilots	Four earlier genuine libsvm linear fits retained in the full log
Foundation model	Two exact prompts on every validation example. One selected prompt on every test example

The ANN search covered AdamW learning rates of 0.0001/0.0003/0.001/0.003, batch sizes of 16/32/64 and convolutional depths of 2/3/4 blocks. The search also covered momentum SGD rates of 0.001/0.01/0.03/0.1, weighted and unweighted BCE, flips enabled or disabled, widths 8/16, dropout 0/0.2 and four configurations using genuine MobileNet transfer learning. These are specific controlled changes and candidates, rather than every possible combination.

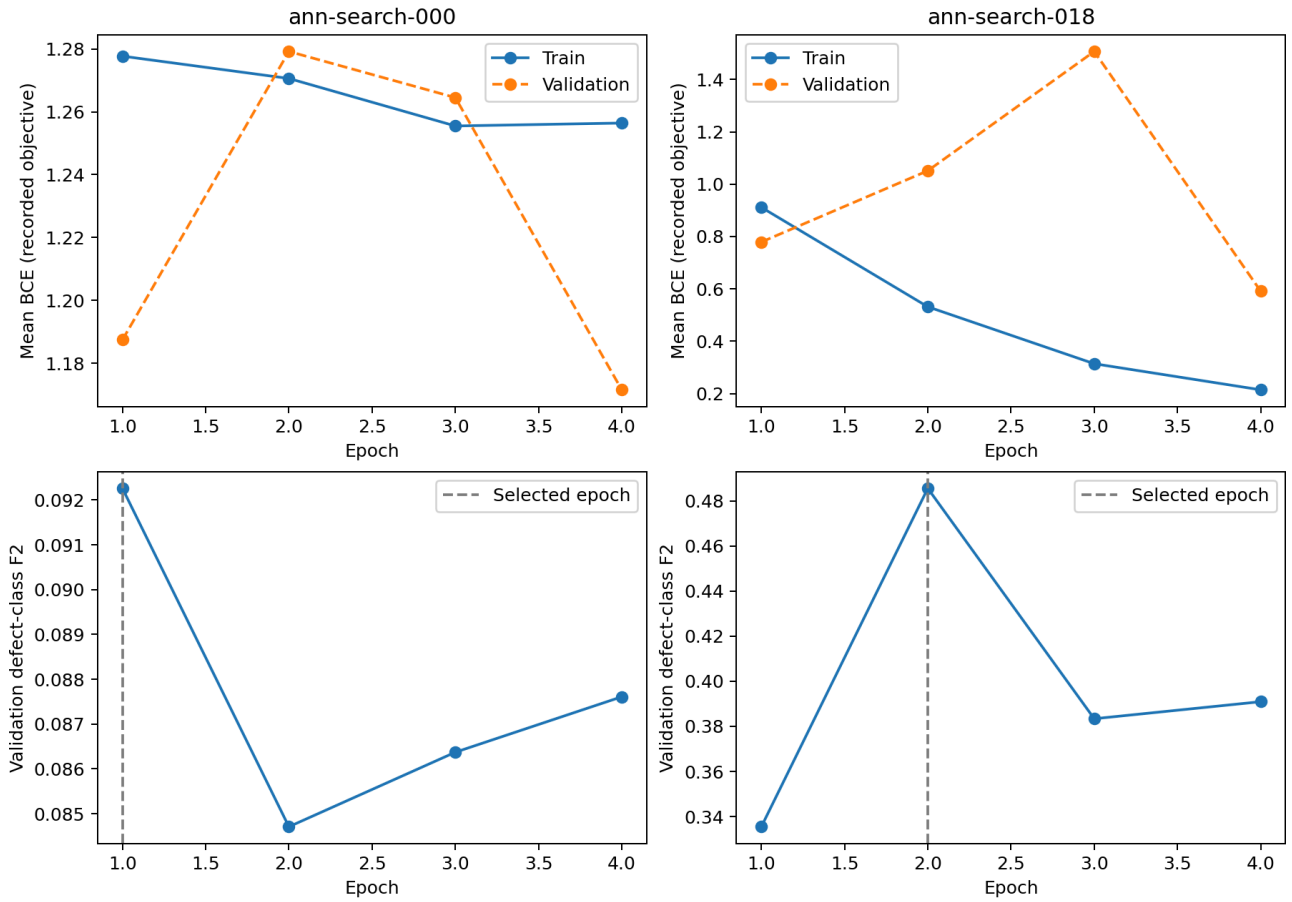
The classical models use either 160-dimensional spatial 4x4 local binary pattern (LBP) histograms or 8,260-dimensional histogram of oriented gradients (HOG) plus LBP features. StandardScaler is fitted only on training data. The main linear fits use LinearSVC with squared hinge loss, tolerance 1e-4, an original dual=auto setting and max_iter 10,000. A numerical change checked only on training data tries dual=True for 2,000 iterations on the remaining HOG+LBP fits with C >=1. Any convergence warning triggers a fresh fit using the full original solver. Validation performance never determines that fallback. RBF SVC uses gamma=scale and a 1,024 MiB kernel cache. C takes values 0.1/1/10, and class_weight is None or balanced. LinearSVC differs mathematically from the earlier pilot models using hinge loss. Final parameters, warnings, iteration counts and every numerical attempt remain in the records.

The ANN screening and refinement took approximately 3.10 CPU elapsed hours, while classical fitting took 2.67 elapsed hours. These are configuration budgets of a similar order, rather than equal counts of floating point operations (FLOPs). Classical feature extraction is charged when first performed and reused afterwards. Training runs shared hardware, so contention limits comparisons of elapsed time.

The complete reports/experiment-log.csv and JSON records retain every configuration, change, hypothesis, validation result, duration, code/data/checkpoint hash and lesson. Runtime resets, interrupted engineering checks and earlier synthetic development tests are labelled separately and do not count as scientific failures. A recovered partial VLM response sequence is reused only after checking the exact source, image, model revision and parser. It is scored only as part of the completed population.

Q3a Learning curves and model development

How the initial and final neural models learned



Learning curves for the initial and final ANN

For the initial model, ann-search-000, training BCE decreased from 1.2777 to 1.2564 and validation BCE decreased from 1.1875 to 1.1716. Epoch 1/4 was selected, with validation F2 of 0.0923 and AP of

0.0188. The checkpoint comes from the best recorded validation epoch, which is not necessarily the last epoch.

For the final model, ann-search-018, training BCE decreased from 0.9116 to 0.2142 and validation BCE decreased from 0.7790 to 0.5912. Epoch 2/4 was selected, with validation F2 of 0.4856 and AP of 0.4991. Again, the saved checkpoint is the best recorded validation epoch rather than automatically the last one.

The initial model's weighted loss changes little over four epochs, and it produces many false alarms. This suggests limited early ability to distinguish defects. Flat or improving training loss alongside weak validation ranking is consistent with insufficient convergence, a shift between source groups or an architecture that dilutes tiny regions. One learning curve cannot establish which mechanism is responsible. Different defect prevalence also affects weighted BCE in training and validation, so their loss gap needs caution. If validation performance worsens while training improves, choosing the final epoch would be unwise.

The final model was selected from measured candidates. Improvement is judged through validation F2 and AP, followed by the held out comparison, rather than through decreasing training loss alone. Matched controls are needed to interpret changes to architecture, optimiser or objective. The combined difference between the initial and final models cannot be attributed to a single cause.

Q3a Comparison on the held out test set

All models use the same 2,996 test patches and decisions fixed on validation data

Model	F2	Precision	Recall	TP/FP/FN	Train (s)	Mean ms
always normal	0.0000	0.0000	0.0000	0/0/96	0.0	N/A
always defective	0.1420	0.0320	1.0000	96/2900/0	0.0	N/A
classical	0.0623	0.2941	0.0521	5/12/91	169.3	36.91
initial ann	0.0000	0.0000	0.0000	0/0/96	841.2	6.46
final ann	0.6883	0.6182	0.7083	68/42/28	625.3	7.44
vlm	0.1391	0.0314	0.9792	94/2900/2	no task fit	764.18

Scored model	AP	ROC-AUC	FPR	P95 total ms
classical	0.1014	0.6127	0.0041	42.18
initial ann	0.0379	0.4927	0.0000	7.15
final ann	0.7185	0.8989	0.0145	9.14

The final ANN has the higher F2 of the ANN and classical model families, supporting the conditional expectation recorded before testing. The baseline that marks every patch as defective shows what happens when every patch is called defective under F2. At least one learned family exceeds that reference, but this alone does not demonstrate suitability for production. No architecture, cutoff, prompt or seed was changed in response to the test results.

Timing was measured on Linux x86_64 using only a CPU, with eight allocated cores. Final inference used one thread and batch size one for every model, after three warmup runs per model. Time for each image includes decoding, preprocessing and prediction, but excludes startup and model loading. VLM parsing and retries are included. These are fresh reconstruction measurements because the first result files were lost during a runtime reset. All outputs for each image were actually recomputed with the same model, threshold and prompt. Other background work overlapped part of VLM inference, so the timings describe this run rather than an isolated benchmark. Earlier confusion counts were used only for cross checking and never replaced inference. Baselines that return a constant output have no meaningful

measured latency. The VLM had no training specific to the task, and the cost of its original pretraining is unknown.

The final ANN's nominal 95% bootstrap interval for F2 using groups is [0.6810035842293907, 0.6976744186046512]. It uses 1,000 resamples, seed 42 and 2 independent test components. With only two origin groups, the resamples contain very few distinct group combinations. The interval therefore describes uncertainty within this dataset and does not provide reliable coverage for a wider production population.

Q3a Error patterns and inspection recommendations

Describing errors after testing without changing the model

Held out origin	N	Positives	F2	Recall	FP / FN
Rollo1A	2436	53	0.6810	0.7170	29 / 15
Rollo3A	560	43	0.6977	0.6977	13 / 13

Positive annotation size	Positive patches	Missed	Recall
<=1% mask area	19	12	0.3684
1% to 5% mask area	40	11	0.7250
>5% mask area	37	5	0.8649

Recall is 36.84% for annotations covering <=1% of a patch, compared with 86.49% for annotations covering >5%. The model misses 12/19 of the small annotated defects. This supports a limitation with tiny regions, although size, origin and texture effects have not been separated. Differences between origins may reflect changes in acquisition or texture. The origin identifiers do not establish physical rolls or manufacturers, and mask area is not a verified measure of severity. Q5 examines three actual mistakes, while the overall error counts retain every test patch.

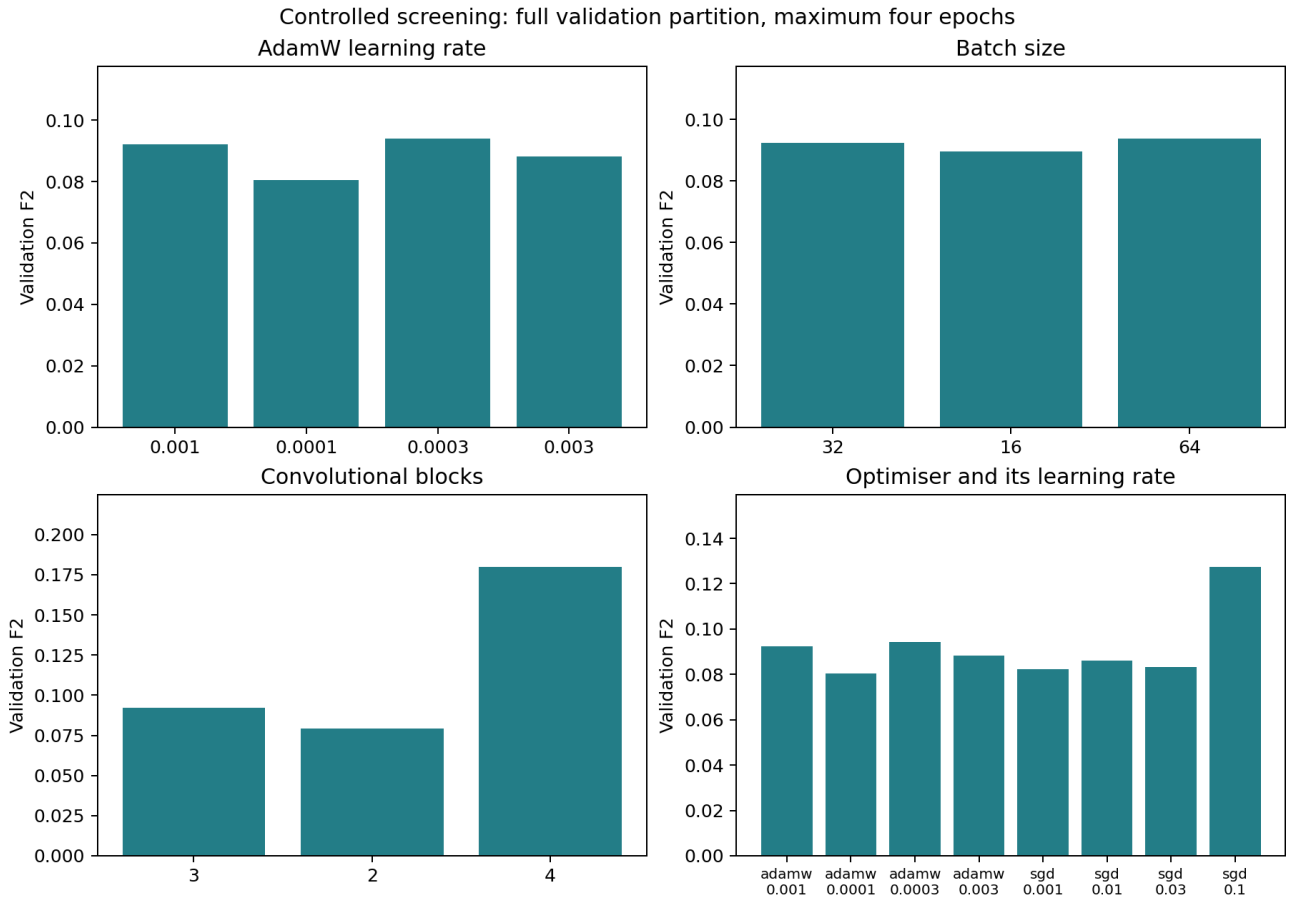
Recommendation	Patches	Defects	Normal
routine inspection	2858	28	2830
manual review	116	46	70
priority hold	22	22	0

The three inspection recommendations use two cutoffs chosen on validation data, low=0.844765 and high=0.998834. The assumed cost is 10 for a defect left in routine inspection, 1 for any manual review and 3 for unnecessarily holding a normal patch. On the held out set, 4.61% of patches are sent for review or hold, while 28 defective patches remain in routine inspection. These costs are coursework assumptions. With rare defects and weak ranking, a policy that sends almost everything to routine inspection can appear numerically cheap. It does not establish operational safety or uncertainty. Ordinary human inspection must continue, and fabric must never be released automatically.

The scores remain uncalibrated, with temperature 1. The held out Brier score is 0.131388 and the validation Brier score is 0.377381. Reliability bins use the same data used for selection, so they do not independently validate confidence. The binary cutoff chosen for F2 and the cutoffs for three inspection recommendations address different stated objectives.

Q3b Effects of learning rate batch size and depth

Measured effects within the declared training budget



Validation results for learning rate batch size and depth

In the AdamW learning rate study, validation F2 was 0.0923 for ann-search-000, 0.0804 for ann-search-001, 0.0941 for ann-search-002 and 0.0881 for ann-search-003. The strongest result in this matched screen of four epochs was ann-search-002.

In the batch size study, validation F2 was 0.0923 for ann-search-000, 0.0895 for ann-search-004 and 0.0939 for ann-search-005. The strongest result in this matched screen of four epochs was ann-search-005.

In the convolutional depth study, validation F2 was 0.0923 for ann-search-000, 0.0793 for ann-search-006 and 0.1799 for ann-search-007. The strongest result in this matched screen of four epochs was ann-search-007.

Learning rate controls update size and affects convergence and stability. A small rate may need more training, while a large rate may cause oscillation. Batch size changes both how gradients are combined and how many updates occur per epoch. Batch size 16 gives about twice as many updates as 32, so using the same epoch count does not isolate gradient noise. Changing depth alters the number of convolutional blocks, the hierarchy of receptive fields, parameter count and class activation map (CAM) resolution. Extra pooling may discard tiny signals. These are findings within a limited experiment, rather than universal settings to recommend.

Q3b Optimisers and variation between training runs

What the optimiser comparison and repeated seeds show

Run	Optimiser	LR	Val F2	Val AP	Best epoch
ann-search-000	adamw	0.001	0.0923	0.0188	1
ann-search-001	adamw	0.0001	0.0804	0.0140	3
ann-search-002	adamw	0.0003	0.0941	0.0165	1
ann-search-003	adamw	0.003	0.0881	0.0153	2
ann-search-008	sgd	0.001	0.0823	0.0180	4
ann-search-009	sgd	0.01	0.0860	0.0150	2
ann-search-010	sgd	0.03	0.0834	0.0162	3
ann-search-011	sgd	0.1	0.1274	0.0333	1

The best matched AdamW result was ann-search-002, with F2 of 0.0941. The best momentum SGD result was ann-search-011, with F2 of 0.1274. Each optimiser was tested over its own learning rate range. The scratch architecture, loss, augmentation, batch size, seed and maximum epoch count stayed the same, while optimiser and learning rate varied together as declared.

AdamW adapts updates using first and second moments, with betas 0.9/0.999 and epsilon 1e-8, and applies decoupled weight decay. SGD uses momentum 0.9 without Nesterov. Both use weight decay 1e-4. SGD may need larger learning rates and longer training. Neither uses a scheduler, so the comparison describes early performance within the available budget, rather than superiority after full convergence. Training examples are shuffled using a seeded generator, with zero data loader workers and no dropped final batch. Full defaults and rules for selecting checkpoints are recorded in docs/training-method.json.

Seed for the selected model	Validation F2
42	0.4856
123	0.3704
2026	0.4527

Across the three selected model seeds, mean validation F2 is 0.4362, sample standard deviation (SD) is 0.0594 and the range is 0.1152. The declared rule treats a range >0.02 as material variation, and this condition is True. All three configurations were fixed before test access, and reconstruction did not change the seed choice. Mean held out F2 is 0.7013, with sample SD of 0.0412. The application continues to use seed 42 regardless of the test ranking.

Q3b Two substantive approaches that failed

Completed experiments and what can reasonably be concluded

svm-search-024 compared with svm-search-036. The hypothesis was that spatial LBP alone could capture irregular weave while keeping the feature vector small.

Validation F2 was 0.0740 for svm-search-024 and 0.1252 for svm-search-036. AP was 0.0148 and 0.0331 respectively. Their TP/FP/FN counts were [38, 2323, 14] and [20, 571, 32].

The controlled change was from 160-dimensional spatial LBP to 8260-dimensional HOG combined with the same LBP. Both used LinearSVC with C=0.1, no class weighting, the same split and a scaler fitted only on training data.

LBP alone ranked defects only about as well as validation prevalence and produced 2323 false alarms. Adding directional gradients increased AP and F2 and reduced false alarms to 571, although recall fell. The LBP histograms lose gradient orientation and boundary information. This controlled comparison supports the value of those added features, but does not establish that every possible LBP variant would fail.

The lesson is to retain HOG plus LBP as a meaningful tuned alternative and report the tradeoff between finding defects and the inspection burden.

ann-search-000 compared with svm-search-036. The hypothesis was that a small CNN trained from scratch with weighted BCE could learn local defects from the approved training images.

Validation F2 was 0.0923 for ann-search-000 and 0.1252 for svm-search-036. AP was 0.0188 and 0.0331 respectively. Their TP/FP/FN counts were [41, 1973, 11] and [20, 571, 32].

This compares whole methods: a weighted scratch CNN and a HOG/LBP linear SVM. Architecture, features and optimisation differ, so the experiment cannot isolate the causal effect of a single factor.

The initial CNN produced 1973 validation false alarms and AP of 0.0188, only slightly above prevalence of 0.0150. Weighted training loss fell from 1.2777 to 1.2564, and the best F2 occurred at epoch one. Within the CPU budget of four epochs, it did not learn a selective detector that transferred across origins. Limited origin diversity, global averaging of tiny defects and insufficient convergence are plausible explanations, but the experiment does not separate their contributions. Different prevalence also affects weighted training and validation losses, so their raw gap alone does not prove overfitting.

The lesson is to test depth and capacity, class weighting, augmentation and genuine pretraining, followed by the declared longer refinement. The small CNN is a valid initial baseline, without any guarantee that the ANN approach will succeed.

Both failures concern substantive feature descriptor or whole method choices. Neither was caused deliberately or counted from a syntax error. The ANN/SVM comparison supports rejecting this CNN approach within the short budget relative to a useful comparator. It does not isolate one causal property of the architecture. All other tested candidates remain in the experiment log.

Q3b Two further unsuccessful attempts

Four distinct measured failures across the study

ann-search-001 compared with ann-search-000. The hypothesis was that reducing AdamW's learning rate from 0.001 to 0.0001 might stabilise training.

Validation F2 was 0.0804 for ann-search-001 and 0.0923 for ann-search-000. AP was 0.0140 and 0.0188 respectively. Their TP/FP/FN counts were [48, 2729, 4] and [41, 1973, 11].

Only the learning rate was changed. The same seed, architecture, batch size, augmentation and weighting were retained, with the same budget of four epochs.

The lower learning rate increased recall but produced 2729 false alarms and reduced F2 and AP. Training loss decreased slowly from 1.3150 to 1.2657, while validation loss rose from 1.0121 to 1.2736. With the same short budget, smaller updates did not improve the ranking of defects. More epochs might change this result. The evidence rejects this tuning attempt within its budget, rather than low learning rates in general.

The lesson is to judge learning rate within a stated optimisation budget, using ranking quality and false alarms as well as the training trajectory.

ann-search-003 compared with ann-search-000. The hypothesis was that increasing AdamW's learning rate from 0.001 to 0.003 might learn useful features faster.

Validation F2 was 0.0881 for ann-search-003 and 0.0923 for ann-search-000. AP was 0.0153 and 0.0188 respectively. Their TP/FP/FN counts were [41, 2077, 11] and [41, 1973, 11].

Only the learning rate changed, with the same tensor values and random number generation rules. NumPy normalisation was independently checked and found identical at the bit level for every possible input gray level. The thread waiting policy affects elapsed time rather than the declared model configuration.

The higher rate kept the same 41 true positives and 11 false negatives but added 104 false alarms. Validation F2 peaked at epoch two and then declined, while validation loss fluctuated. This supports rejecting faster updates within the short study. It does not establish a universal stability threshold or a statistically significant difference with only one validation origin.

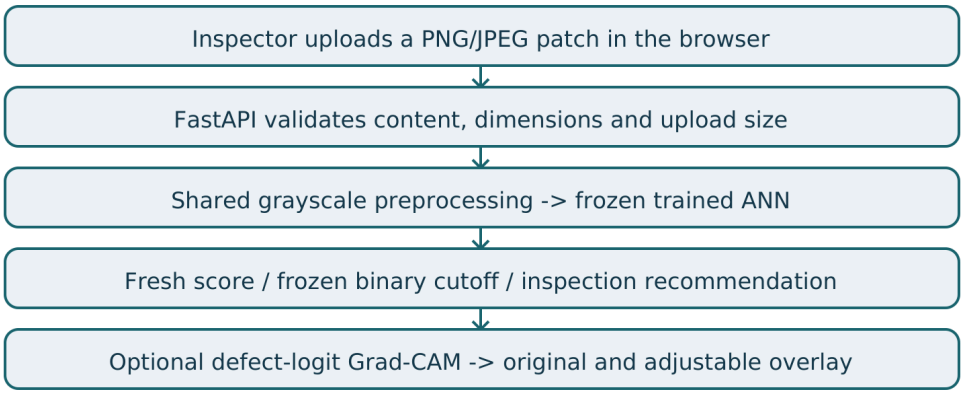
The lesson is that larger updates do not automatically help. The record needs to show the actual specificity and learning trajectory.

These diagnoses use training and validation evidence recorded before final testing. A low learning rate failed within this budget but is not inherently invalid, and more epochs might change its result. A higher rate added false alarms at similar recall, but four epochs cannot establish the eventual optimum. Execution interruptions remain separately recorded and do not count as failed scientific approaches.

Q4 The web application and its tests

The selected trained ANN answers fresh requests

The inspector uploads a PNG or JPEG patch in the browser. FastAPI checks its contents, dimensions and size, then applies the shared grayscale preprocessing and fixed trained ANN. The response contains a fresh score, a decision using the frozen binary cutoff and an inspection recommendation. An optional Grad-CAM explanation of the defect logit is displayed alongside the original image with adjustable overlay opacity.



Application request and explanation flow

FastAPI loads the selected checkpoint once and verifies its SHA-256 checksum, preprocessing and provenance. It runs the model for each prediction or explanation request. The /health endpoint reports readiness, and /predict and /explain accept multipart image uploads. A lock protects model hooks and gradients when requests overlap. Missing or corrupt weights return status 503, with no random or prepared prediction as a fallback. Uploaded pixels are neither retained nor sent to an external VLM.

The interface displays the original image and real overlay, opacity controls, the meaning of the uncalibrated score, cutoff, class and three possible inspection recommendations. It also shows the checkpoint version, measured response time and limitations. Uploads must be single image, 8 bit L/RGB, 256x256 PNG or JPEG files of at most 5 MiB. The application decodes image contents rather than trusting the filename or MIME type. Highlights support inspection and do not certify defect boundaries.

Executed check group	Result
Real HTTP validation/parity	13 checks passed
Browser interaction	6 checks passed

Executed check group	Result
Normal image content	Grayscale PNG, RGB PNG and JPEG match direct checkpoint inference
Invalid inputs	Empty/malformed/wrong size/alpha/paletted/oversize/invalid opacity rejected
Materially different inputs	Blank and random noise produce finite scores. No reliable OOD detector is claimed

Q5a Three methods for explaining predictions

What each explanation method shows and what it cannot establish

Method	What it explains / cost	Important limitations
Grad-CAM [4]	Convolutional features of a particular architecture. One forward and backward pass for the defect logit	Coarse maps that depend on the target and layer. Global averaging and ReLU suppress information. Not causal or certified boundaries.
Integrated Gradients [5]	Input attribution relative to a baseline. Many gradient evaluations along a path	Reference choice changes attribution. A black/mean fabric is unnatural. Pixel noise and interactions can mislead.
LIME [6]	Local surrogate that works across models using perturbed superpixels. Many forward calls	Superpixel/perturbation/sampling choices. Local fidelity and instability. Cutting periodic weave can make images outside the training distribution.

Grad-CAM was selected because the ANN has convolutional feature maps and the inspector needs a quick spatial cue beside the original patch. It targets the defect logit even when the model predicts normal. It averages gradients to weight feature channels, applies ReLU, aligns the map to 256x256 using bilinear interpolation and normalises by its maximum. A genuinely zero attribution remains zero, so the interface does not create an artificial hotspot.

For the custom CNN, Grad-CAM uses the last block's ReLU before pooling. This retains twice the spatial resolution of the final pooled map. MobileNet uses its final feature block. Comparing a map with the publisher's mask measures spatial alignment with the annotation. Testing faithfulness requires sensitivity checks. Neither measure explains the manufacturing cause. The original image, score and fixed cutoff remain visible so that an attractive heatmap cannot replace visual judgement.

A useful next study would compare different reference baselines and perturbations, and test randomisation of model parameters across more independent origins. These would be new studies declared in advance, rather than changes made after seeing the submission's test results.

Q5b Checking whether the explanations are useful

Equal area occlusion controls and genuine prediction errors

The explanation sample was declared in advance. It selects up to nine cases each from true positives, false negatives, false positives and true negatives (TP/FN/FP/TN), preferring different source images and then filename order. The resulting sample has 36 cases, including 18 genuine errors. All 2,996 test examples still contribute to the main metrics. In 13/36 cases, the CAM is nonzero and removing the most highlighted region produces a positive logit drop larger than both controls. This does not establish faithfulness for every case.

Three selections are used for occlusion, each covering 10% of the pixels: pixels with the highest CAM values, pixels with the lowest CAM values and a random mask generated with seed 42. Selected pixels are replaced by their image's mean value for each channel. The measured response is the original defect logit minus the perturbed logit. A positive drop means less defect evidence, while a negative drop means the score increased. A positive top region drop larger than both controls supports only local sensitivity. Replacing pixels with their mean can produce implausible texture, and ranking tied or zero CAM values can be arbitrary.

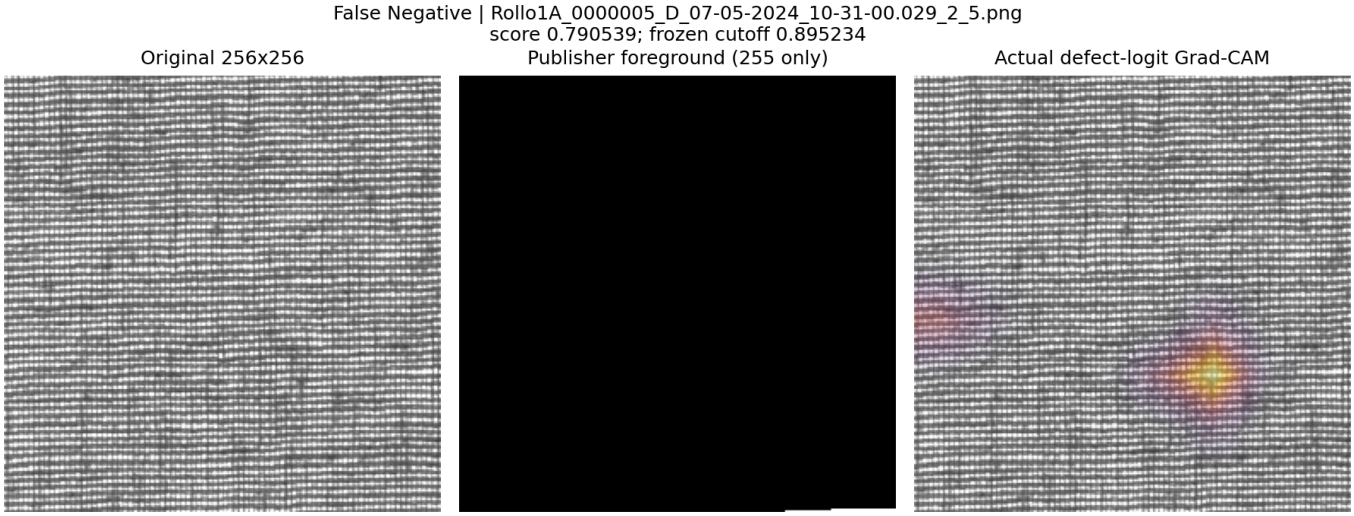
Measured case	Label/pred	Top drop	Low drop	Random drop
Rollo1A_0000001_D_07-05-2024_10-30-55.085_0_1.png	1/1	5.092145	5.195260	-0.283907
Rollo1A_0000002_D_07-05-2024_10-30-56.352_3_0.png	1/1	6.521844	1.033612	0.865592

Among positive annotated cases with a nondegenerate map, the attribution peak falls inside the mask in 2/18 cases. Each case also compares the share of attribution inside the mask with the mask's share of image area under a uniform attribution map. Negative patches have empty masks, so these pointing measures and measures of attribution mass are undefined rather than assigned artificial zeros. Agreement with a mask shows correspondence, but does not prove that the network used the true cause of the defect.

Three genuinely incorrect predictions from this sample are analysed below, including both error types where available. Numerical checks can disagree with a visually appealing map, so unfavourable results remain in the analysis. Explanations and error diagnoses are not used to change the fixed weights, thresholds or prompts.

Q5b Error 1 A missed defect

Rollo1A_0000005_D_07-05-2024_10-31-00.029_2_5.png
Publisher origin Rollo1A



Original image annotation and Grad-CAM for the first missed defect

This patch is genuinely defective under the publisher's label 1, but the model predicts 0. Its score is 0.790539, below the cutoff of 0.895234 chosen on validation data. The annotation covers 103 pixels, or 0.157% of the patch. Occluding top, low and random regions of equal area changes the defect logit by +1.591740, +2.704260 and +0.498417 respectively. The highlighted region therefore does not show stronger sensitivity than both controls. Filling occluded pixels with the mean may create unnatural weave. The map neither identifies a causal production fault nor certifies the defect boundary.

The original image mainly shows repeating horizontal weave. The publisher's tiny annotated region lies at the bottom edge towards the right, while the brightest Grad-CAM region sits higher in the lower right interior. The attribution peak misses the mask. Only 0.00134% of total attribution falls inside it, below the 0.15717% reference for uniform attribution. This is consistent with poor attention to a tiny boundary defect, although it does not establish downsampling as the cause. The score of 0.790539 remains below the fixed 0.895234 cutoff. Removing the top region reduces the logit, but removing the region with low attribution reduces it more, by 2.704260 compared with 1.591740. The explanation therefore fails the check against both controls. A future study could test features that preserve boundary detail on new validation groups, without changing this evaluated model.

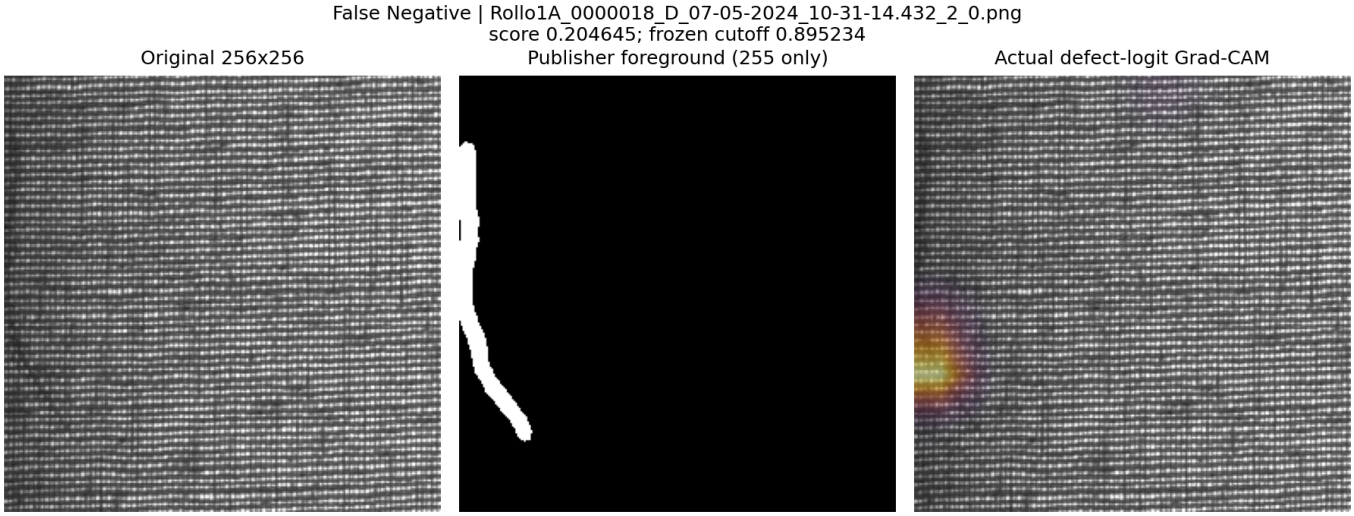
The mask contains foreground pixels, but the pixel with the highest attribution lies outside them. Attribution mass inside the mask is 0.000013, compared with a reference for uniform attribution of 0.001572. A pointing check for defects has no target when the mask is empty.

The original defect logit is 1.328180, and the occluded area is 10.00%. The difference between the top and random drops is 1.093323, and the difference between the top and low drops is -1.112520. The zero CAM flag is False. The case JSON retains the full numerical precision.

The publisher's annotation is the ground truth for this benchmark, but it does not independently verify the mechanism of a fabric fault. The exact original, mask and overlay can be reproduced from the cited manifest row and saved checkpoint.

Q5b Error 2 A second missed defect

Rollo1A_0000018_D_07-05-2024_10-31-14.432_2_0.png
Publisher origin Rollo1A



Original image annotation and Grad-CAM for the second missed defect

This defective patch has ground truth label 1 but prediction 0. The score is 0.204645, below the fixed cutoff of 0.895234 chosen on validation data. The publisher's mask covers 1764 pixels, or 2.692%. Occluding top, low and random regions of equal area changes the logit by -0.427480, -0.421962 and -0.641229 respectively. The highlighted region does not show stronger supporting sensitivity than both controls. Filling occluded pixels with the mean may create unnatural weave, and the map does not establish a production cause or certified boundary.

The annotation follows a narrow irregular region along the left edge, where the original image shows subtle variation in weave and intensity. The coarse CAM brightens the lower left area and partly overlaps the annotation. The mask contains 21.3191% of attribution while occupying 2.69165% of the image, which exceeds the uniform reference. However, the single attribution peak remains outside the mask of exactly white pixels. Overlap alone does not explain the low score of 0.204645 or certify a boundary. All three occlusions increase the defect logit, giving negative drops. Top deletion gives -0.427480, compared with -0.421962 for deletion of pixels with low attribution and -0.641229 for random deletion. The perturbation therefore does not validate the highlight as selectively supporting the defect decision. Mean filling may itself disturb the weave, and neither production cause nor annotation severity is established.

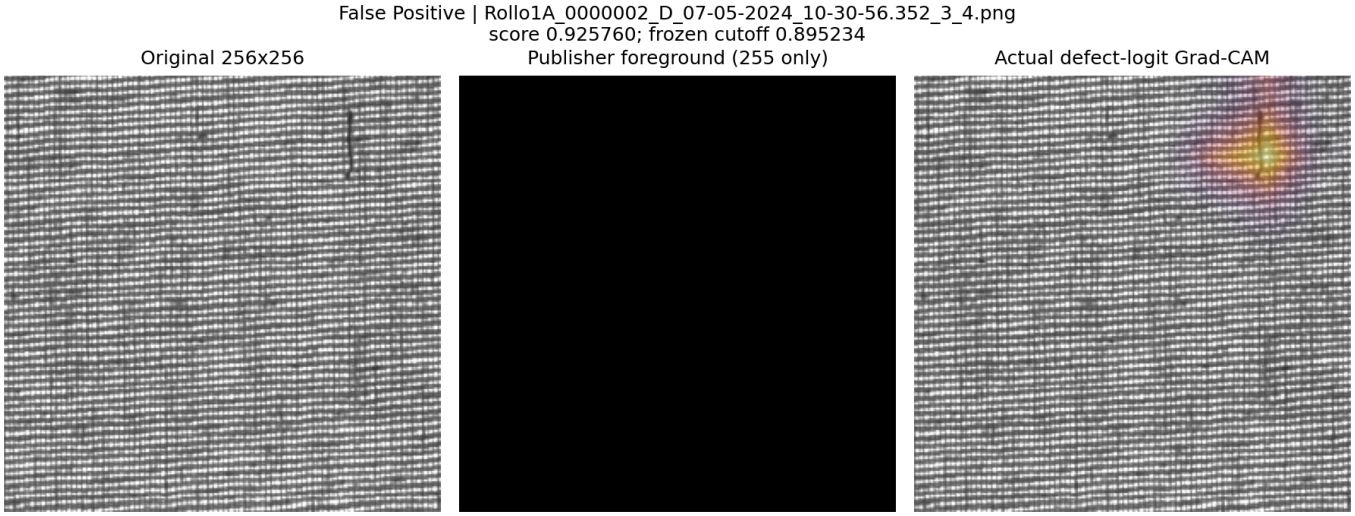
The mask contains the foreground, but the attribution peak lies outside it. Attribution mass inside the mask is 0.213191, compared with a reference for uniform attribution of 0.026917. A pointing check for defects has no target when the mask is empty.

The original defect logit is -1.357512, and the occluded area is 10.00%. The difference between the top and random drops is 0.213750, and the difference between the top and low drops is -0.005518. The zero CAM flag is False. Full precision remains in the case JSON.

The publisher's annotation supplies the ground truth for this benchmark without independently verifying the mechanism of a fabric fault. The exact original, mask and overlay can be reproduced from the cited manifest row and saved checkpoint.

Q5b Error 3 A false alarm

Rollo1A_0000002_D_07-05-2024_10-30-56.352_3_4.png
Publisher origin Rollo1A



Original image annotation and Grad-CAM for the false alarm

This normal patch has ground truth label 0, but the model predicts 1. Its score of 0.925760 exceeds the fixed validation cutoff of 0.895234. The publisher's mask contains 0 defect pixels, or 0.000%. Occluding top, low and random regions of equal area changes the defect logit by +2.779529, +3.000914 and -0.592691 respectively. The highlighted region does not show stronger sensitivity than both controls. Mean filling may create unnatural weave, and the map does not establish a causal production fault or certified defect boundary.

The original image has a regular weave and a short, narrow, dark vertical feature near the upper right. Grad-CAM peaks nearby, although the publisher's mask is empty and the patch is explicitly labelled normal. A score of 0.925760 above the fixed 0.895234 cutoff makes this a genuine false alarm under the publisher's target. The highlight is consistent with sensitivity to a distinctive texture feature, but does not prove an unannotated defect or an incorrect label. With an empty mask, overlap and pointing hit scores are undefined. They should not be interpreted as zero localisation scores. Top deletion lowers the logit by 2.779529, but deletion of pixels with low attribution lowers it more, by 3.000914. Random deletion raises it. The explanation again fails the check against both controls. Independent inspector annotation would be needed before changing any label.

The mask is empty, so the location of the attribution peak, attribution mass in the mask and reference for uniform attribution are all N/A. There is no annotated defect for a pointing check.

The original defect logit is 2.523316, and the occluded area is 10.00%. The difference between the top and random drops is 3.372220, and the difference between the top and low drops is -0.221385. The zero CAM flag is False. Full precision is retained in the case JSON.

The publisher's annotation defines the ground truth for this benchmark without independently identifying a mechanism of a fabric failure. The exact original, mask and overlay can be reproduced from the cited manifest row and saved checkpoint.

Q6a Evaluating a local vision language model

SmolVLM-256M-Instruct with a prompt chosen on validation data

The assessed generative model is HuggingFaceTB/SmolVLM-256M-Instruct [7], licensed under Apache 2.0 and fixed at revision 7e3e67edbbed1bf9888184d9df282b700a323964. This compact model for images and text can make direct binary patch decisions and runs on the available CPU. It is a pretrained comparator, while the trained ANN remains the coursework model.

The run was configured with transformers 4.57.1 and genuine downloaded safetensors, with trust_remote_code=False. It runs in bfloat16 on one CPU thread, with greedy decoding, a maximum of 24 new tokens, batch size one and one retry after a parsing or inference failure. The original 256x256 pixels are preserved without tiling or upscaling. At this size, the processor uses 16 matching visual tokens, a compression that may lose tiny defects. There is no paid API, although hardware, electricity, local compute time and the initial download still have costs.

Prompt 0: Inspect this fabric patch for a visible manufacturing defect. Reply with only one label: defective or defect_free.

Across all 3472 validation patches, Prompt 0 achieved F2 of 0.0707, precision of 0.0150 and recall of 1.0000. There were 74 unresolved outputs, or 2.13%.

Prompt 1: Classify this fabric patch. Look for a local disruption, hole, stain or anomalous yarn rather than ordinary repeating weave or lighting. Reply with only one label: defective or defect_free.

Across all 3472 validation patches, Prompt 1 achieved F2 of 0.0707, precision of 0.0150 and recall of 1.0000. There were 23 unresolved outputs, or 0.66%.

Prompt index 1 was selected using validation F2, then lower failure rate and finally smaller index. Neither prompt correctly classified any normal validation patch, so their high recall does not show useful discrimination. The exact prompt, configuration, parser and model revision were fixed before any test score was available. The saved validation response sequence was completed without changing the protocol, and raw responses are retained. Partial results never determined prompt selection.

Only image pixels and the prompt are provided to the model. PNG metadata is removed. Labels, masks, revealing filenames, source groups, ANN scores and hints derived from test data are excluded. The parser requires one exact defective/defect_free token or a single canonical decision sentence. Ambiguous prose that remains after the fixed retry is retained as unresolved and counted as an error. Responses, attempts, input and output tokens, and timings are saved for every example.

Q6b Comparison deployment conditions and a possible extension

What the measured results support

Evidence	Trained ANN	Generative VLM
Test population	2,996 same frozen patches	2,996 same frozen patches
F2 for the defect class	0.6883	0.1391
Precision / recall	0.6182 / 0.7083	0.0314 / 0.9792
FP / FN	42 / 28	2900 / 2
Mean total latency	7.44 ms	764.18 ms
Unresolved output	0	48
Ranking / probability	Continuous uncalibrated model score	Label only. No invented AP/AUC/probability
Explanation	Measured Grad-CAM for the defect logit	Generated label is not a faithful causal explanation

The compact ANN has a small, fixed interface, fewer dependencies and a reproducible local checkpoint. It still needs representative labelled origins and may miss tiny defects or unfamiliar textures. The VLM can be prompted without weight fitting specific to the task and might help with related descriptions. However, it requires more dependencies, download and compute resources, compresses visual detail and can return invalidly formatted outputs. Its pretraining provenance cannot establish that the benchmark was entirely absent from its training data. Neither model's results guarantee performance in an unseen factory.

Deployment decisions should consider measured F2, tolerance for missed defects, inspection capacity and latency. The model's architectural reputation is not evidence of suitability. Local ANN serving keeps uploaded factory images within the application process, and the study also runs the VLM locally. A hosted VLM would introduce external service availability, data governance, pricing and reproducibility considerations. Those costs were not incurred here, and no numerical values are invented for them.

A realistic extension would turn verified inspection fields into a short structured note. Its inputs would be defect status confirmed by an inspector, patch ID and observed location. The assistant would need to quote those confirmed fields, avoid inventing severity or root cause, and request human review. A new study with separate origin groups for testing could compare usefulness and unsupported claims. The present Q6 remains a direct classification experiment on the complete shared population, without claiming an exemption for a related task.

Appendix A Reproducibility and verification

Verification records and reproduction procedure

Report findings are based on actual runs using the publisher's complete RAW-FABRID release. The retained evidence includes model checkpoints, configurations, training histories, predictions, explanations, timings and raw VLM responses. ChatGPT/Codex assisted with execution and analysis. The report's claims are supported by these saved results. Earlier synthetic development checks are kept separate from the assessment metrics, and engineering interruptions and runtime resets are recorded separately from unsuccessful scientific approaches.

The generative model assessed in Q6 is a genuine local SmolVLM-256M-Instruct fixed to a publisher revision. Its inference from images, prompts, parser and failures are recorded separately from the use of ChatGPT/Codex as a productivity assistant. Simulated VLM results do not satisfy Q6.

A runtime reset during a pause in the work lost the first completed final-evaluation files. The same fixed protocol was physically recomputed to reconstruct them. There was no retraining or change to the model, threshold, prompt or split. Delivered outputs and timings are genuine reconstruction measurements, and earlier aggregate counts serve only as cross checks. The recovery record discloses two physical computations of one benchmark, rather than independent repeated validation.

Verification was completed with 52 passing tests, successful lint and compilation checks, and successful checks through actual HTTP requests and browser interactions. All notebook cells are executed with rich outputs in separate IPython processes because the hosted environment blocks the standard Jupyter socket transport. The report was rendered and visually checked. Supporting records are notebook-verification.json, docs/requirements-matrix.md and verification.json.

For a quick check after extracting the project, install the pinned CPU environment with the test/vlm/notebooks extras and run python -m pytest -q. Launch uvicorn with FABRIC_BUNDLE=artifacts/serving-bundle.json and FABRIC_CPU_THREADS=1, then upload authentic data/demo cases. A full reproduction downloads the verified release and pinned pretrained weights and uses a fresh output folder. Existing fixed results must never be overwritten or silently evaluated again.

Appendix B References and supporting evidence

Primary sources and reproducible project files

[1] Perez-Llorens, R., Albero-Albero, T. and Silvestre-Blanes, J. (2026). RAW-FABRID V1. Mendeley Data. DOI 10.17632/db6g85xsyg.1. https://data.mendeley.com/datasets/db6g85xsyg/1

[2] Howard, A. et al. (2019). Searching for MobileNetV3. https://arxiv.org/abs/1905.02244

[3] PyTorch torchvision. MobileNetV3 Small ImageNet weights and model documentation. https://docs.pytorch.org/vision/stable/models/generated/torchvision.models.mobilenet_v3_small.html

[4] Selvaraju, R. R. et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. https://arxiv.org/abs/1610.02391

[5] Sundararajan, M., Taly, A. and Yan, Q. (2017). Axiomatic Attribution for Deep Networks. https://arxiv.org/abs/1703.01365

[6] Ribeiro, M. T., Singh, S. and Guestrin, C. (2016). Why Should I Trust You?: Explaining the Predictions of Any Classifier. https://arxiv.org/abs/1602.04938

[7] Hugging Face. SmolVLM-256M-Instruct model card. Pinned commit recorded in this report. https://huggingface.co/HuggingFaceTB/SmolVLM-256M-Instruct

[8] Scikit-learn. LinearSVC and SVC documentation. https://scikit-learn.org/stable/modules/generated/sklearn.svm.LinearSVC.html

Claim / output	Exact evidence
Dataset, split, labels	data/manifests/split.json / split.csv / data-audit.json
EDA and modelling constraints	docs/eda-statistics.json / eda-modelling-findings.json
Every training run	experiments/*.json reports/experiment-log.csv
Architecture and curves	reports/*-architecture.csv reports/figures/
Four failures	docs/failed-approaches.json
Frozen protocol / reconstruction	artifacts/final-evaluation-manifest.json reports/final-evaluation/test-access.json
Final scores for each example	reports/final-evaluation/{initial_ann,final_ann,classical,vlm}.json
Three genuine XAI errors	docs/three-xai-errors.json reports/final-evaluation/xai/
Application inference at request time	reports/application-verification/results.json / browser-results.json
Tests and notebooks	reports/execution-checks.json / notebook-verification.json

Appendix C. Generative AI Declaration

OpenAI ChatGPT and Codex were used as supporting tools during this coursework. AI assistance included clarifying requirements, exploring possible approaches, supporting code development and debugging, assisting with dataset preparation, experiments and testing, analysing results, preparing notebooks, and improving the clarity of the report.

SmolVLM was used separately as the generative AI model evaluated in Question 6, with its implementation and results documented in that section.

I remain responsible for understanding and verifying all submitted material, including code, experimental results, interpretations, references and conclusions. This includes any material developed with AI assistance. I also remain responsible for explaining and defending the submitted work during the individual live presentation.